

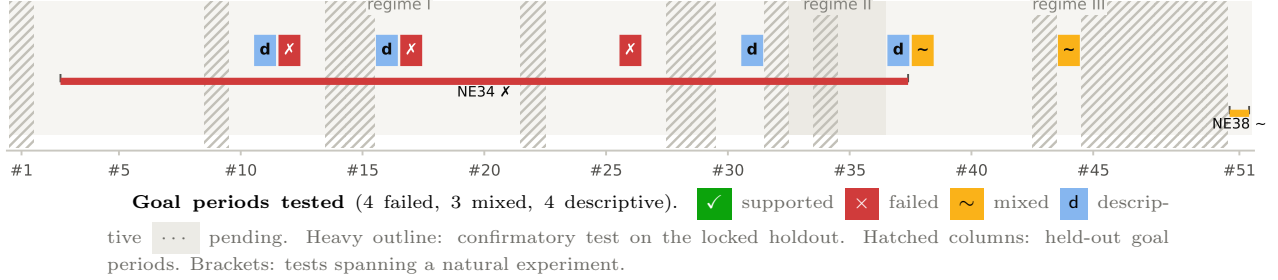
# H10: Goals are Legendre pushes: a swarm's response to an assigned goal is predictable from its unforced fluctuations

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1/5 Usefulness 1.5/5

**Bottom line.** Goals are quenches toward a common target, not Legendre pushes: free-week fluctuations do not predict who follows, in both embedding models.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



## WHAT HAPPENED

The question is whether a swarm's unforced fluctuations predict how it lines up behind an assigned goal. The state is each agent's 30-min statement embedding, projected on the goal direction  $\hat{g}$ . Three free  $\rightarrow$  assigned pairs exist (#11 $\rightarrow$ #12, #16 $\rightarrow$ #17, #37 $\rightarrow$ #38). In each pair one field  $\lambda$  fits the mean shift, and the model predicts the rest. Alignment rises 1.4–3.1 free-week SDs, outside linear response. **The core test fails:** fluctuations do not predict who moves ( $r = -0.37, -0.69, +0.20$ ; Stouffer  $p = 0.89$ ). **Round 1b** uses shared goal vectors, a second embedding model, dedupe and style residualization. The test fails in all 8 input configurations. Variance along  $\hat{g}$  grows, and transverse variance holds. In a new native test a reassigned goal moves one agent 10–20 SDs, beyond any point its unforced content visits. An agent-set goal moves agents, but again not in fluctuation order. The holdout is frozen and has not run.

## PHYSICAL MODEL

Model 11: mean-field  $O(n)$  in  $n = 32$  whitened embedding dimensions.  $\mathbf{s}_i$  are agent  $i$ 's unit statement vectors.  $y = \hat{\mathbf{z}} \cdot \hat{\mathbf{g}}$  is the alignment with the goal field direction.  $x_i$  is its 30-min window mean, with free-week mean  $\mu_i$ , noise-corrected variance  $\kappa_{2,i}$  and third cumulant  $\kappa_{3,i}$ . The goal is a linear field  $\lambda = \beta h$  along  $\hat{g}$ .  $K$  is the cumulant generating function,  $J_0$  the mean-field coupling,  $g$  the loop gain and  $C_F$  the free-week window covariance.

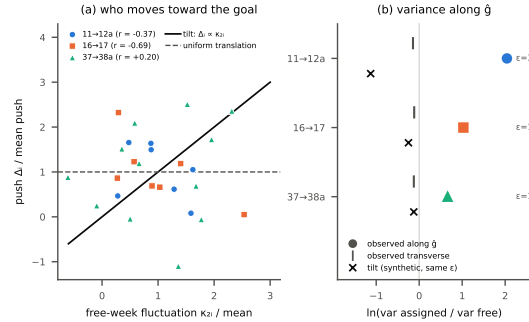
$$P_A(\mathbf{s}) \propto P_F(\mathbf{s}) e^{\lambda \sum_i \hat{\mathbf{g}} \cdot \mathbf{s}_i}, \quad G = F - \lambda m$$

$$K_A(t) = K_F(t + \lambda) - K_F(\lambda) \Rightarrow \kappa_n^A = K_F^{(n)}(\lambda)$$

$$\Delta \mu_i = \lambda \kappa_{2,i} + \frac{\lambda^2}{2} \kappa_{3,i}, \quad \kappa_{2,i}^A = \kappa_{2,i} + \lambda \kappa_{3,i}$$

$$\chi_{\text{swarm}} = \chi_0 / (1 - \beta J_0 \chi_0), \quad g = \beta J_0 \chi_0 = 1 - V_{\text{indep}} / V$$

$$\Delta \mu = \lambda C_F \hat{g}$$



**Observables.** (a) Each agent's push toward the assigned goal against its free-week fluctuation along  $\hat{g}$ , both divided by the pair mean. The tilt predicts the solid line, uniform translation the dashed one. The regime-I pairs slope the wrong way. (b) The change in variance along  $\hat{g}$  (colored), along transverse directions (bars), and for an exact tilt of a bounded mean-field swarm at the same push ( $\times$ , synthetic). The goal disperses the swarm along  $\hat{g}$ ; it does not narrow it.

## SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The promise was to read a swarm's steerability from its unsteered fluctuations. That promise does not hold here. A goal change is a large kick (median about 2.5 SDs), not a small field. The agents that wander most are not the agents that follow most. An assigned goal acts as a common target and a new collective mode, not as a gentle tilt of existing preferences. Do not forecast compliance from pre-goal behavior: the fluctuation–response correlation runs from  $-0.69$  to  $+0.20$ . The instrument (goal-direction projection, noise-corrected variance) is cheap on any agent log. The fluctuation–response physics it tests is a curiosity in this village.

H10-goals-are-legendre-pushes · generated 2026-10-07 17:49 by infra/summaries/build\_summaries.py · card hypotheses/H10-goals-are-legendre-pushes/README.md · ratings: RE-C1 re-evaluation agent (suggested)

## RESULTS VS PREDICTIONS

| Prediction                      | Round 1 $\rightarrow$ 1b (bge / gte)                                                                   | Verdict      |
|---------------------------------|--------------------------------------------------------------------------------------------------------|--------------|
| P1 fluctuations order who moves | $r$ $-0.37, -0.69, +0.20 \rightarrow$ same / $\times$<br>$-0.53, -0.49, -0.09$ ; 8/8 configs fail      | $\times$     |
| Push $\Delta\bar{\mu} > 0$      | 3/3 pairs; 19/19 kickoffs $\rightarrow$ same $\checkmark$<br>in all configs                            | $\checkmark$ |
| P2 variance by the tilt         | untestable ( $\varepsilon > 0.34$ ); gte n/a<br>#37 $\rightarrow$ #38 would pass ( $\varepsilon$ 0.69) | n/a          |
| Cross-kickoff man $> 0$         | Spear- $-0.56 \rightarrow -0.56 / -0.59$                                                               | $\times$     |
| G44 assigned vs own-choice room | #best $\varepsilon$ 0.97 / 1.55; #rest none; $\sim$ gap +0.17                                          | $\sim$       |
| NE38 one agent re-signed        | $\varepsilon$ 10 / 20; post mean beyond the pre support $\sim$ (tilt $\times$ )                        | $\times$     |
| G26 agent-set goal              | push $\varepsilon \approx 1$ ; P1 $r$ $-0.14 / -0.63$                                                  | $\times$     |

## FAITHFULNESS SCORECARD

|                        |   |                   |   |
|------------------------|---|-------------------|---|
| A mapping              | 1 | F identifiability | 1 |
| B assumptions          | 1 | G ground truth    | 1 |
| C adequacy             | 0 | H comparative     | 0 |
| D unfitted predictions | 0 | I transfer        | 0 |
| E interventional       | 0 |                   |   |

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

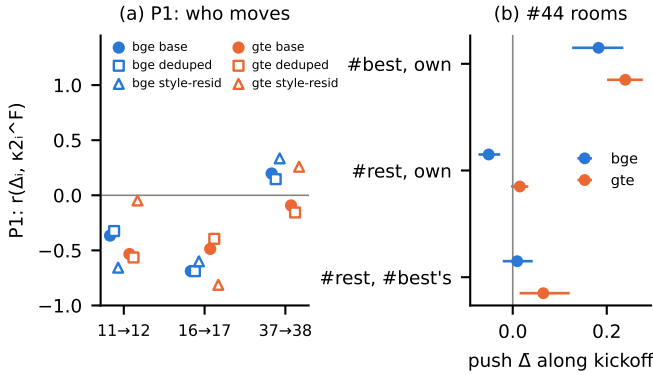
## CAVEATS

$\hat{g}$  is a text-embedding proxy for the field. gte shrinks the regime-III push and makes its variance tilt-compatible, so the regime-III result depends on the model. Regime I has  $N = 7$ . NE38's concurrent control is one incumbent, which drifts 1–3 SDs along its own goal. The #44 assigned room has 4 agents and three days. The natives are exploratory. #22 $\rightarrow$ #23 and #31 $\rightarrow$ #32 remain blind.

## ROUND 2

Re-map goals as a two-state (on/off-goal) occupancy. Test field against contagion onsets (H10-R1/R2). Run `confirm.py` only after Vivian's sign-off.

- **Essence.** How does a new goal propagate: as a field acting on everyone at once, or as contagion from early adopters?
- **H10-R1** Adoption onsets after a kickoff: a field predicts simultaneous onsets; contagion predicts onsets ordered by exposure to early adopters.
- **H10-R2** Goals act on a two-state on-goal/off-goal occupancy (HH130).
- **H10-R3** During the week, goal-specific content relaxes back toward family-prior directions (E4).



**Second figure.** (a) P1 per pair under every round-1b input (blue bge, orange gte; filled base, squares deduped, triangles style-residualized). The regime-I pairs stay negative. (b) #44 native: the assigned room moves along its own kickoff, and the self-chosen room does not (90% CIs).

## GOAL PERIODS IN DETAIL

G11 d 1b: descriptive (both models)  
G12  $\times$  1b: failed (both models)  
G16 d 1b: descriptive (both models)  
G17  $\times$  1b: failed (both models)  
G26  $\times$  failed (round 1b native; both models)  
G31 d 1b: descriptive (both models)  
G37 d 1b: descriptive (both models)  
G38  $\sim$  1b: mixed (bge mixed, gte failed)  
G44  $\sim$  mixed (round 1b native; model-dependent)  
NE34  $\times$  1b: failed (both models)  
+1 more periods: see the card.